EDWARD DAVIES

Digital Engineering Specialist | Digital Twins | Synthetic Data | Computer Vision

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British Citizen • Full Clean UK Driving Licence

PROFESSIONAL PROFILE

Digital Engineering Specialist with hands-on experience developing and delivering digital twin, synthetic data and AI solutions within Jaguar Land Rover's manufacturing environment. Sole technical owner and project manager of an end-to-end synthetic data pipeline, integrating large-scale vehicle CAD, OpenUSD, photorealistic digital twins and Python-based NVIDIA Omniverse automation to generate annotated datasets for computer vision model development and validation. Experienced in taking innovative digital engineering concepts from proof-of-concept through technical development and evaluation, combining strong technical delivery with project leadership, stakeholder communication and customer-facing experience.

TECHNICAL & PERSONAL SKILLS

Digital Engineering & 3D: NVIDIA Omniverse, OpenUSD/USD, Deltagen, Blender, Dassault 3DEXPERIENCE, CAD-to-USD workflows, USD scene graphs/prims, Digital twins, Photorealistic materials
Synthetic Data & Computer Vision: Synthetic data generation, Automated bounding boxes, COCO JSON, JSONL, CSV, RF-DETR, GCP AutoML, Model training/evaluation, Image augmentation, CLAHE, Colour balancing
Programming & Data: Python (pandas, NumPy, Scikit-learn, Matplotlib, NLP libraries), Omniverse extension scripting, SQL, Java, MATLAB, JavaScript, HTML, CSS, PHP
Core Competencies: Technical project ownership, project management, cross-functional leadership & team delegation, presentations, proof-of-concept development, adaptability & rapid tech adoption, creative problem solving

PROFESSIONAL EXPERIENCE

JAGUAR LAND ROVER

National Automotive Innovation Centre, University of Warwick

Digital Graduate Scheme, Digital Specialist - Industrial Operations - *Sep 2024 – 2026 - Present*

Pipeline Ownership:
Sole technical owner and project manager of an end-to-end synthetic-data pipeline for AI-based vehicle build inspection, designed to create training data for unreleased vehicle programmes ahead of physical production.
CAD & 3D Workflows:
Process large-scale vehicle CAD in Deltagen, export to OpenUSD/USD, and integrate vehicle assets in NVIDIA Omniverse with photorealistic digital twins of JLR inspection tunnels created in Blender.
Automation & Scripting:
Develop Python extensions in NVIDIA Omniverse to traverse and manipulate complex USD scene structures, control vehicle variants, and automate synthetic image generation.
Dataset Generation:
Automate annotation and dataset export, generating thousands of images with bounding boxes for targeted vehicle components and exporting labelled data to JSONL, CSV, and COCO JSON formats.

AI Model Training:
Trained and evaluated computer-vision models in GCP using AutoML; tested CLAHE, noise, colour balancing, and image augmentation. Transitioned to an internal RF-DETR pipeline for local training and benchmarking against curated test datasets.

3D Factory Automation:
Developed an Omniverse automation workflow that recursively traverses production-area folders and USD assets to apply reusable photorealistic materials to factory zones (robots, grippers, body-handling equipment).

Leadership & Stakeholder Management:
Led delivery independently, coordinated contributing colleagues, and regularly presented technical progress and updates to senior management.

NORTHUMBRIA UNIVERSITY | **AI Research Intern**

Faculty of Engineering, Newcastle upon Tyne • May 2023 – Aug 2023

Conducted research into causal discovery and causal inference alongside a PhD candidate.
Reviewed newly published academic literature, datasets, and codebases.
Built a structured repository of modern datasets and code implementations to support ongoing university research.

EARLY & ADDITIONAL EMPLOYMENT

GoDaddy (Nottingham) – Support and Sales *(Feb 2024 – July 2024)*

Aldi (Derby) – Warehouse Associate *(Oct 2023 – Feb 2024)*

Northumbria University (Newcastle Upon Tyne) – Enterprise Placement Year *(Sep 2021 – Jun 2022)*

Amazon (East Midlands Airport) – Warehouse Associate *(May 2020 – Sep 2020)*

Sitel Telethon (Newcastle upon Tyne) – Customer Service Advisor *(Nov 2019)*

Topshop Topman (Newcastle upon Tyne) – Customer Assistant *(Feb 2019 – Jun 2019)*

Tesco (Long Eaton, Nottingham) – Customer Assistant & Small Team Leader *(Nov 2016 – Aug 2017)*

EDUCATION & QUALIFICATIONS

NORTHUMBRIA UNIVERSITY | **BSc (Hons) Computer Science with Artificial Intelligence**

First-Class Honours (1st) with Year in Industry • 2019 – 2023

Individual Computing Project (Dissertation): Investigated optimization techniques for fake news detection using AI models.
Key Modules: Machine Learning & Computer Vision, Internet of Things, Artificial Intelligence & Robotics, Team Project and Professionalism.
Leadership: Regularly stepped up as team leader for group assessments, managing workload allocation, scheduling meetings, and driving project completion.